MANU DWIVEDI

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SUMMARY

Director of Engineering, AI/ML R&D at Tiami Networks, leading a team of about 20 engineers. I build products that give telecom operators new revenue from the network they already run: radar sensing on live 5G base stations with no new radios or vendor changes, rapidly deployable passive radar for counter-UAS, RF digital twins that plan sensing before deployment, zero-day threat detection for 5G SA and private networks, and world models that sell autonomy to robot and drone fleets as a network service. Counter-UAS sensors sold into approximately every major US and EU operator; PolyRAN trialled publicly with Vodafone and AirSpan.

LEADERSHIP

Tiami Networks

Director of Engineering, AI/ML R&D *Oct 2025 to Present*

- Lead about 20 engineers across machine learning, signal processing, RF, embedded systems, and robotics. Own the roadmap and delivery from lab to licensed deployment in carrier and defense networks.
- Raised about **$6M** for sensing and ISAC research and productization. Executive engagement with Verizon, T-Mobile, SoftBank, AirSpan, Deutsche Telekom, and Corning. Public trials and demos with Vodafone, AirSpan, Deutsche Telekom, NVIDIA, and Northeastern University.

OPERATOR PROGRAMS

PolyRAN: Vendor-Agnostic Radar Sensing on Live 5G RAN **Tiami Networks**

2024 to Present

- Turns existing base stations into a wide-area radar with no new radios, antennas, or changes to the RAN vendor's software. A sensing dApp reads uplink reference signals the gNB already schedules (SRS, PUSCH DMRS) through a vendor-agnostic O-RAN E3 service model (E3SM), and a radar foundation model turns them into detections and tracks.
- Processing chain: coherent processing intervals, delay-Doppler correlation (CAF), CFAR detection, and tracking, served to an operator UI over HTTP. Needs no dedicated radar waveform and no extra spectrum.
- Runs on a server beside the DU; integrated with OpenAirInterface, OCUDU, and NVIDIA Aerial, with optional O-RAN fronthaul 7.2x ingest when the DU cannot export data.
- **Trialled publicly with Vodafone** at its Málaga R&D lab on live 5G with Open RAN radios, detecting objects and people without disrupting calls, messages, or data. Also trialled with **AirSpan** and tier-2 operators in Asia and the EU. Covered by the Vodafone newsroom, Light Reading, Mobile World Live, and Telecompaper.

PolyEdge and AirRadar: Rapidly Deployable Passive Radar **Tiami Networks**

2023 to Present

- Standalone counter-UAS sensor for sites where the RAN cannot be modified. It reconstructs the existing 5G downlink (SSB, PBCH, CRS), or broadcast TV for AirRadar, as its reference and cancels direct-path clutter (Wiener-Hopf, ECA), so it needs no transmitter and no spectrum license. Shipped as IP-protected containers with hardware-fingerprint licensing (Docker, Helm, OpenShift) to major US and EU operator sites.
- Small drones detected at about **1 km** on public and private 5G (below 400 ft, up to 15 m/s, radar cross-section about 0.01 m^2); 5G-connected drones identified at 95% in clutter; light aircraft tracked to about **25 km** on broadcast TV. Tracks fuse with Remote-ID and ADS-B and reach command systems over SAPIENT, ASTERIX, FAAD C2, and TAK.

Agentic RF Planner: Digital Twins for ISAC **Tiami Networks**

2024 to Present

- City-scale RF digital twin from map and 3D building geometry. Vision-language agents assign building materials and clutter from street imagery; 3GPP TR 38.901 propagation with sector and tilt configuration plans coverage and radar sensing on one scene, with Cesium and 3D exports for operator review. Used to site every Tiami deployment.
- Signal-strength prediction field-validated against commercial UE telemetry in North America, Europe, and Asia (n71, n3, n5, n25, n41, n77, n78, CBRS; NSA and SA; multiple vendors). Representative n41 SA case: **4.22 dB** MAE over 17,324 UE samples.

CoreDiff: Zero-Day Threat Detection for 5G SA and Private Networks **Tiami Networks; PIN-Cyber**

2025 to Present

- Structures the network into a graph of which elements interact, and how, over time. A model trained only on normal behavior scores how unusual each interaction is, so new attack types are caught without signatures. Alarms are calibrated to a set false-alarm rate per hour and map to MITRE ATT&CK. Evaluated with red-team attack runs on a network emulator plus denial-of-service and data-exfiltration scenarios.
- At the 1% anomalous tail: 5G SA about **94% F1**, time to detect cut from 20 to 5 minutes. Private networks about **98.9% F1**, mean time to detect cut from 120 to 0.6 seconds, across 13 ATT&CK techniques.

CoreDiffWFM: Autonomy for Robots and Drones as a Network Service **Tiami Networks**
2025 to Present

- Robots and drones that lack autonomy (no map, no global position, poor sensing) receive it from the network. The world model fuses ISAC and PolyEdge detections, network positioning, fixed cameras, and each agent's own sensors into one shared picture, and returns goals, holds, and keep-out zones for the agent's own navigation. Agents fill the picture where infrastructure coverage is poor.
- **Indoor:** a ground robot with no map or position is localized from fixed cameras and lidar (offline runs) before any goal is sent. **Outdoor:** drone and ground-robot fleets over city geometry; one agent's hazard replans every affected mission.
- The RF planner and a ray-traced physics model predict serving cell, line of sight, and uplink SINR at each time step along each route, explaining network behavior for an agent that must hold its link and choosing upload points that finish sooner on the existing RAN. RAN path validated with an OCUDU gNB, OAI NR-UE, and E3 dApp (PUSCH DMRS SNR 35.6 and 40.6 dB).
- Results: a predicted 1.000 m move observed as 0.980 m on a robot testbed (within three sigma). In simulation, three drones avoided a helicopter and cut fleet completion from 152 to 125 seconds; ground robots finished 6 of 6 missions around a blockage versus 3 of 6.

ARCHES: End-to-End Counter-UAS Response **Tiami Networks**
2025 to Present

- Closes the loop from detection to response. A radar cue hands off to an interceptor drone that tracks the target with onboard vision (YOLOv8) and Remote-ID, working with commercial drones without a vendor SDK. A staged PX4 safety process takes flight control from simulation to real flight on NVIDIA Jetson, with a ground station coordinating several drones.

RF Pattern-of-Life **Tiami Networks**
2025 to Present

- Identifies and locates known and unknown emitters (Wi-Fi, drone control links, 5G NR) from raw RF without decoding them: channelization, burst detection and fingerprinting, and classification by link role. Four-antenna phase-coherent angle of arrival and 5G PRS positioning. Won over **$1M** in follow-on R&D funding.

EARLIER EXPERIENCE

Tiami Networks
Principal Embedded ML Engineer *Dec 2024 to Oct 2025*

- Led sensing proofs of concept with Tier-1 mobile operators, MVNOs, a physical security firm, and the US Army. Uplink gNB pedestrian tracking with NVIDIA and Northeastern University, and ISAC fall detection with Deutsche Telekom and Hubraum (MWC 2025). Pedestrian and traffic monitoring for the City of Elk Grove at 99.6% accuracy from 250 m to 1.3 km.

Tiami Networks
Sr. Embedded ML Engineer and AI FPGA Intern *Sept 2023 to Dec 2024*

- Took 5G and LTE sensing from software-radio prototypes to production on Intel and FPGA hardware; Open RAN sensing for NTIA NOFO 2; Arria10 and RFSoC edge sensing demos at MWC 2024. Closed-loop core and RAN assurance with 3GPP Rel-16 NWDAF won the Deutsche Telekom T-Challenge 2024 award (EUR 25,000). Cross-modal radar and camera models (patent pending).

Michigan State University and IBM OpenPower
Research *2020 to 2023*

- Generative-AI acceleration on low-power FPGAs with a 45% compute reduction; open-ISA processor accelerators with about 4x ALU throughput. Graduate Office Fellowship ($7,500).

PUBLICATIONS AND PATENTS

- M. Dwivedi et al., "5G-Based Passive Radar Sensing for Human Activity Recognition using Deep Learning," *IEEE Radar Conference*, 2024.
- M. Dwivedi et al., "Person Presence Detection using 5G Synchronization Signals on SDR Testbed," *IEEE MILCOM* (demo), 2024.
- Two U.S. patents pending: indoor and outdoor RF activity sensing; cross-modal radar and camera sensing.

TECHNICAL SKILLS

Open RAN : O-RAN E3 and E3SM dApps, OpenAirInterface, srsRAN, OCUDU, NVIDIA Aerial, fronthaul 7.2x, 3GPP NWDAF
ISAC : Passive and bistatic radar, delay-Doppler, CFAR, clutter cancellation, angle of arrival, PRS positioning, DVB-T and ATSC
RF / C2 : USRP, bladeRF, GNU Radio, RFSoC; SAPIENT, ASTERIX, FAAD C2, TAK; Docker, Kubernetes, OpenShift
ML : PyTorch, radar foundation models, diffusion models, vision-language agents, CUDA, TensorRT, NVIDIA Jetson
Languages : Python, C/C++, CUDA C, MATLAB, Verilog/SystemVerilog, Embedded C

EDUCATION

- **Michigan State University**, M.S. Electrical and Computer Engineering, GPA 3.8
- **SRM Institute of Science and Technology**, B.Tech. Electronics and Communication Engineering, GPA 3.8